

Data-Centric Multimodal Pavement Distress Detection Using Intensity–Range Imagery

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GitHub Repository:

<https://github.com/qoa10/independent-study-icmr-2026>



Presentation Roadmap

1. Motivation / Problem

2. TxDOT Intensity–Range Dataset

3. Proposed Method

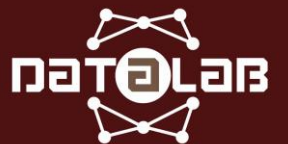
4. Experimental Setup

5. Results

6. Conclusion / What I Tried and Lessons Learned



Motivation / Problem



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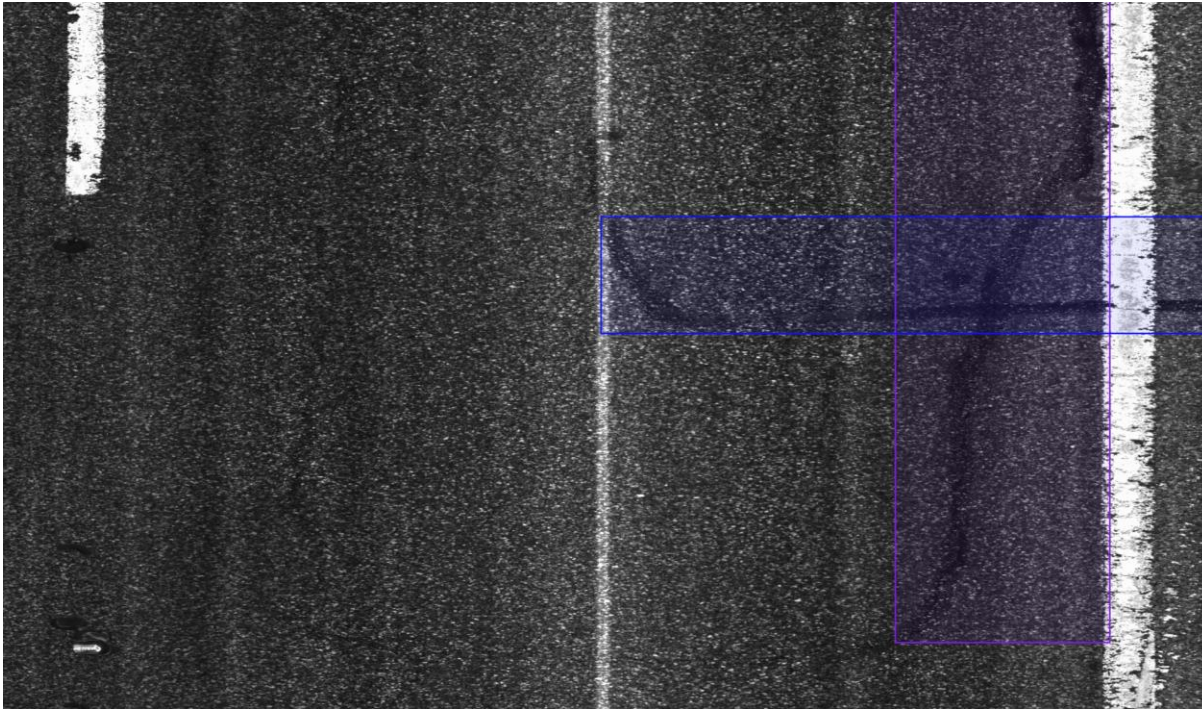
Motivation: Why Intensity–Range Multimodal Detection?

- Pavement distress detection is important for large-scale roadway condition monitoring.
- 2D intensity images capture surface appearance, texture, and visual contrast.
- 3D range images capture surface geometry, depth variation, and structural cues.
- Different distress classes may rely on different modalities.
- The key problem is how to combine intensity and range without losing class-critical information.



Example: Paired Intensity and Range Images

2D Intensity Image



3D Range Image

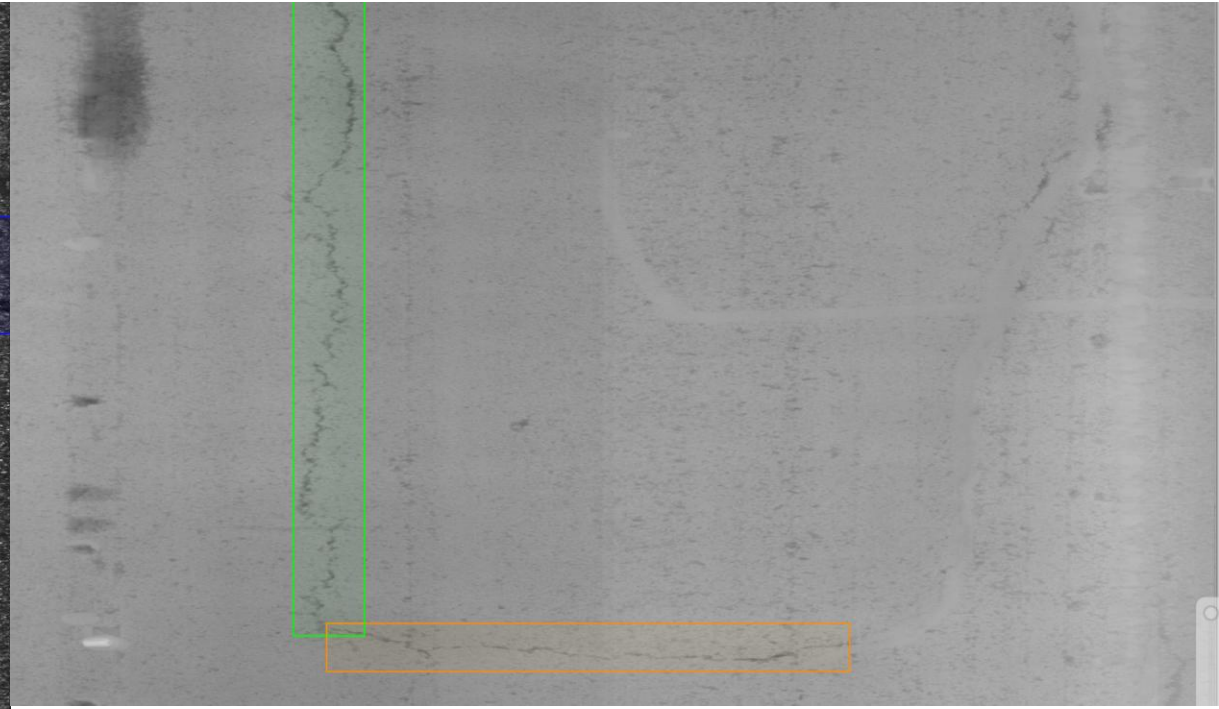


Figure 1: Paired intensity and range images provide complementary 2D appearance and 3D surface cues.

Research Objective

Main Goal

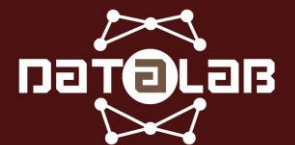
- Improve pavement distress detection by better leveraging complementary information from 2D intensity and 3D range imagery.

Objectives

- 1. Analyze class-wise modality preference between intensity and range.
- 2. Design a split-backbone YOLOv5s model to preserve modality-specific features.
- 3. Develop class- and modality-aware augmentation for paired 2D–3D inputs.
- 4. Evaluate the proposed method against early-fusion and single-modality baselines.

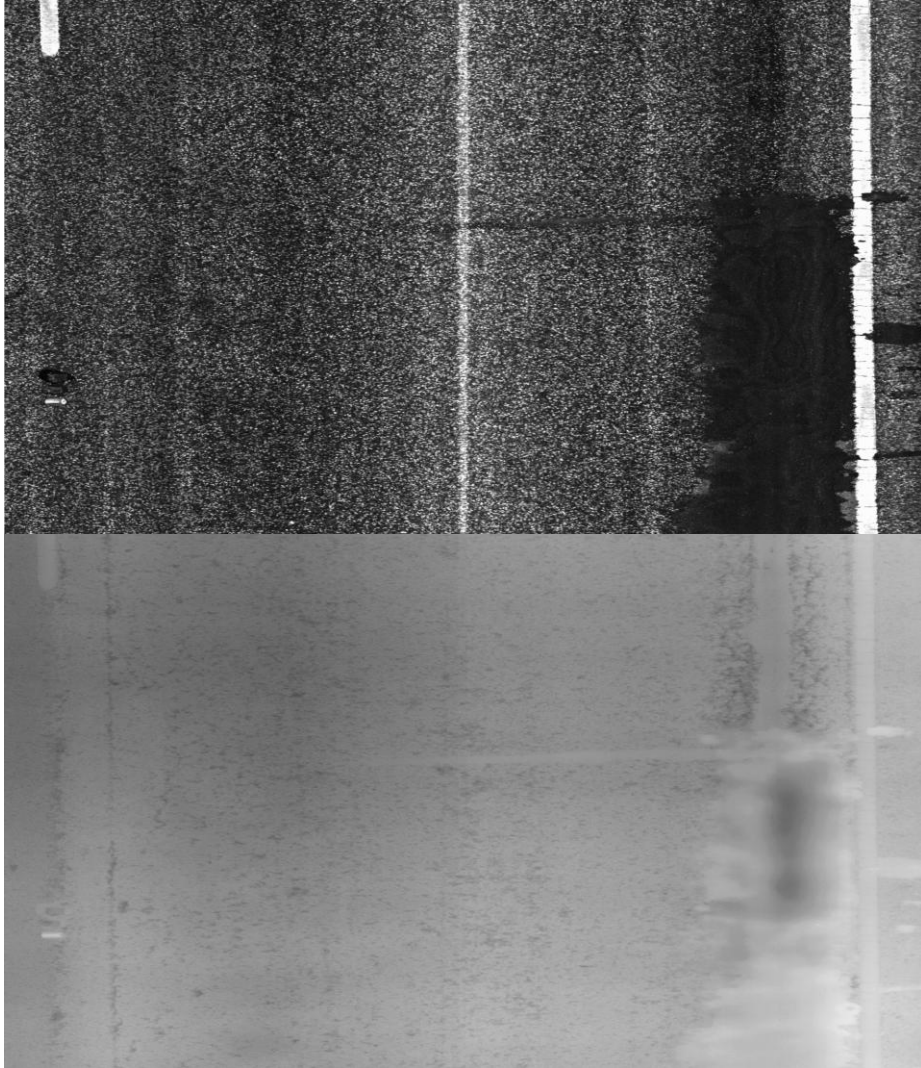


TxDOT Intensity- Range Dataset



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Dataset Overview

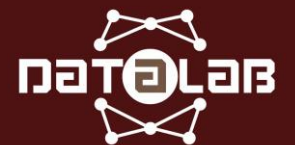


Dataset Source

- TxDOT 2D/3D Asphalt Concrete Pavement (ACP) dataset
- Collected by a pavement inspection vehicle
- Each sample contains spatially aligned intensity and range images

Image Modalities

- 2D intensity images: camera-based pavement surface appearance
- 3D range images: laser-based pavement surface measurements
- Aligned 2D/3D images are tiled into 1536×900 patches for detection annotation and model training.



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Figure 2: Paired 2D intensity and 3D range images from the TxDOT ACP dataset.

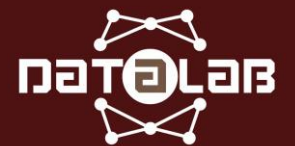
Class distribution

- The TxDOT ACP subset contains nine pavement distress classes.
- The class distribution is highly imbalanced.
- Frequent classes include sealed longitudinal, sealed transverse, longitudinal, and lane longitudinal.
- Rare classes include pothole, block, and alligator.
- Class imbalance makes multimodal detection more challenging, especially for rare distress types.

Table 1: Class distribution of the TxDOT ACP dataset.

Class	# Images	# Instances
Transverse	1,183	1,673
Joint	369	488
Sealed transverse	1,506	2,101
Longitudinal	1,369	1,853
Lane longitudinal	1,383	1,487
Sealed longitudinal	2,411	4,025
Block	293	296
Alligator	678	878
Pothole	115	154
Total	6,936	12,945

Proposed Method



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Proposed Multimodal Detection Workflow

- Input: paired 2D intensity and 3D range images
- Shared geometric augmentation preserves 2D–3D alignment
- Modality-specific pixel augmentation handles intensity and range differently
- Split-backbone YOLOv5s keeps modality features separated in the backbone
- Neck/head performs late fusion and outputs class, box, and confidence

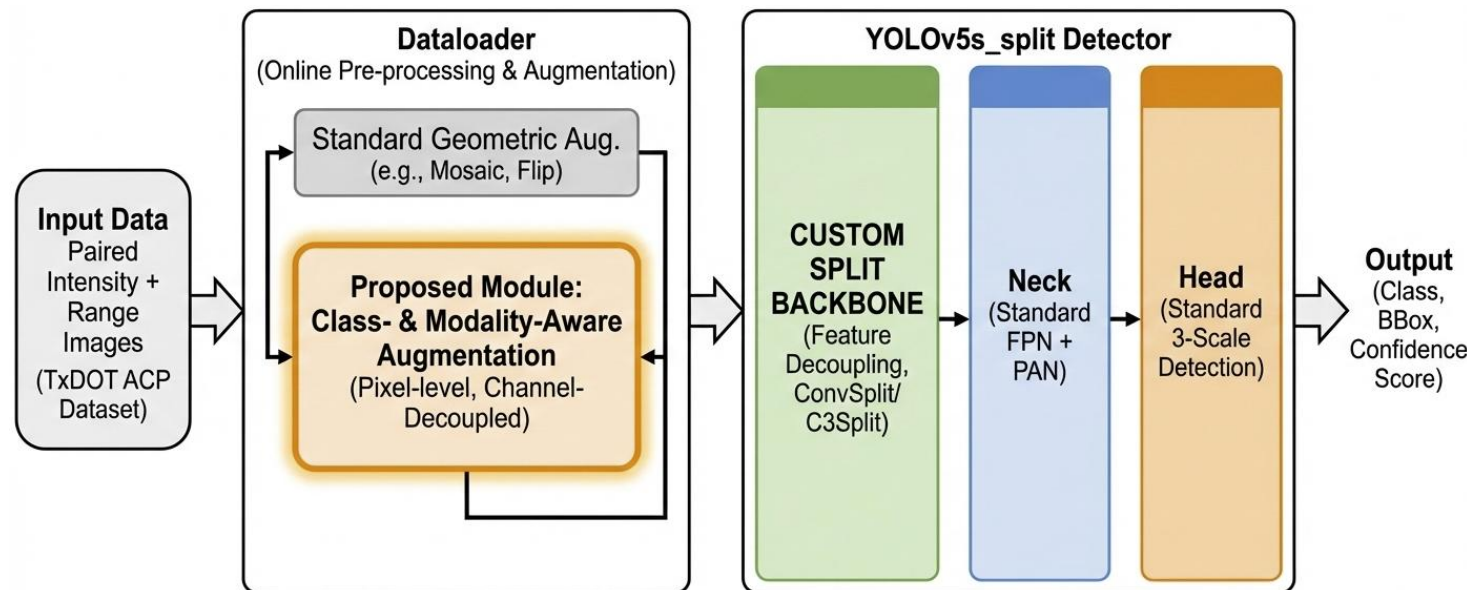
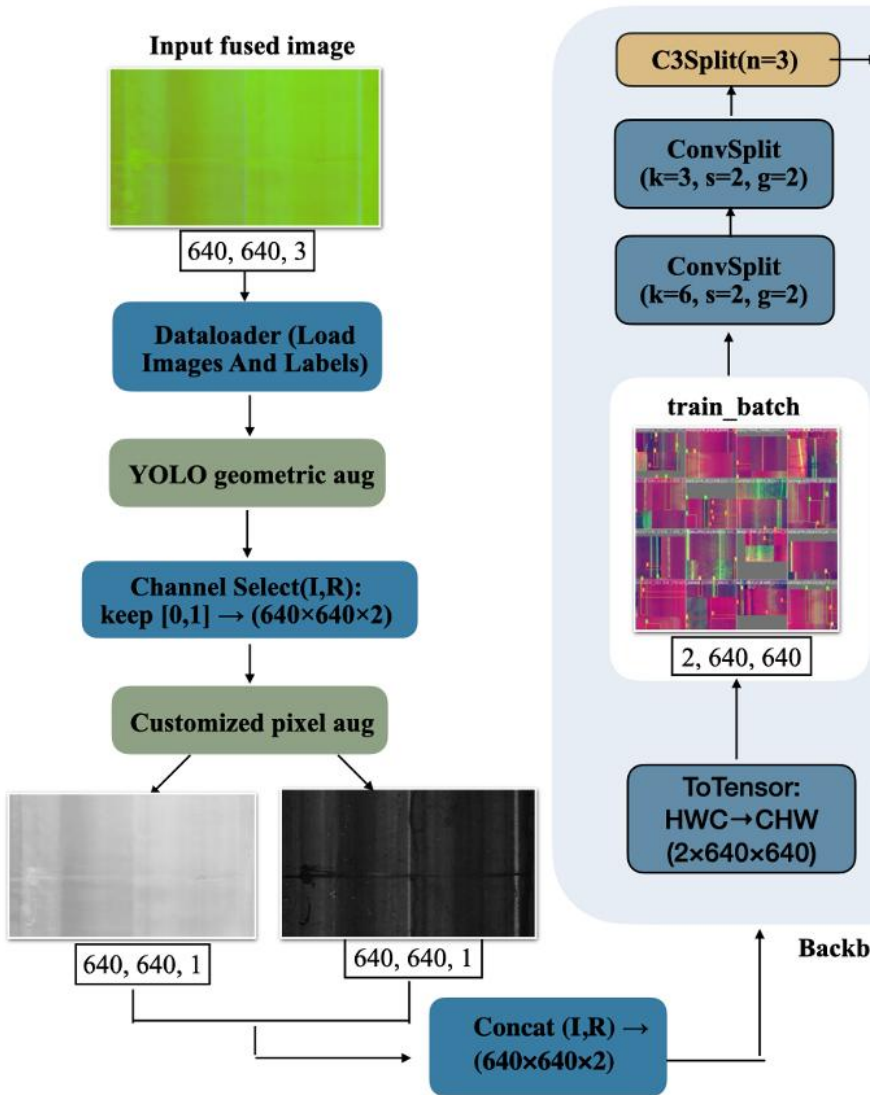


Figure 4: Proposed intensity–range detection workflow with modality-aware augmentation and split-backbone YOLOv5s.

Step 1 — Dual-Channel Data Pipeline



- Input image stores intensity, range, and a dummy channel
- Keep only intensity and range channels: [0, 1]
- Shared geometric augmentation preserves 2D–3D alignment
- Pixel-level augmentation is applied separately to each modality
- Final model input: $640 \times 640 \times 2$ tensor

Figure 4: Dual-channel data pipeline for paired intensity-range inputs.

Step 2 — Split-Backbone YOLOv5s

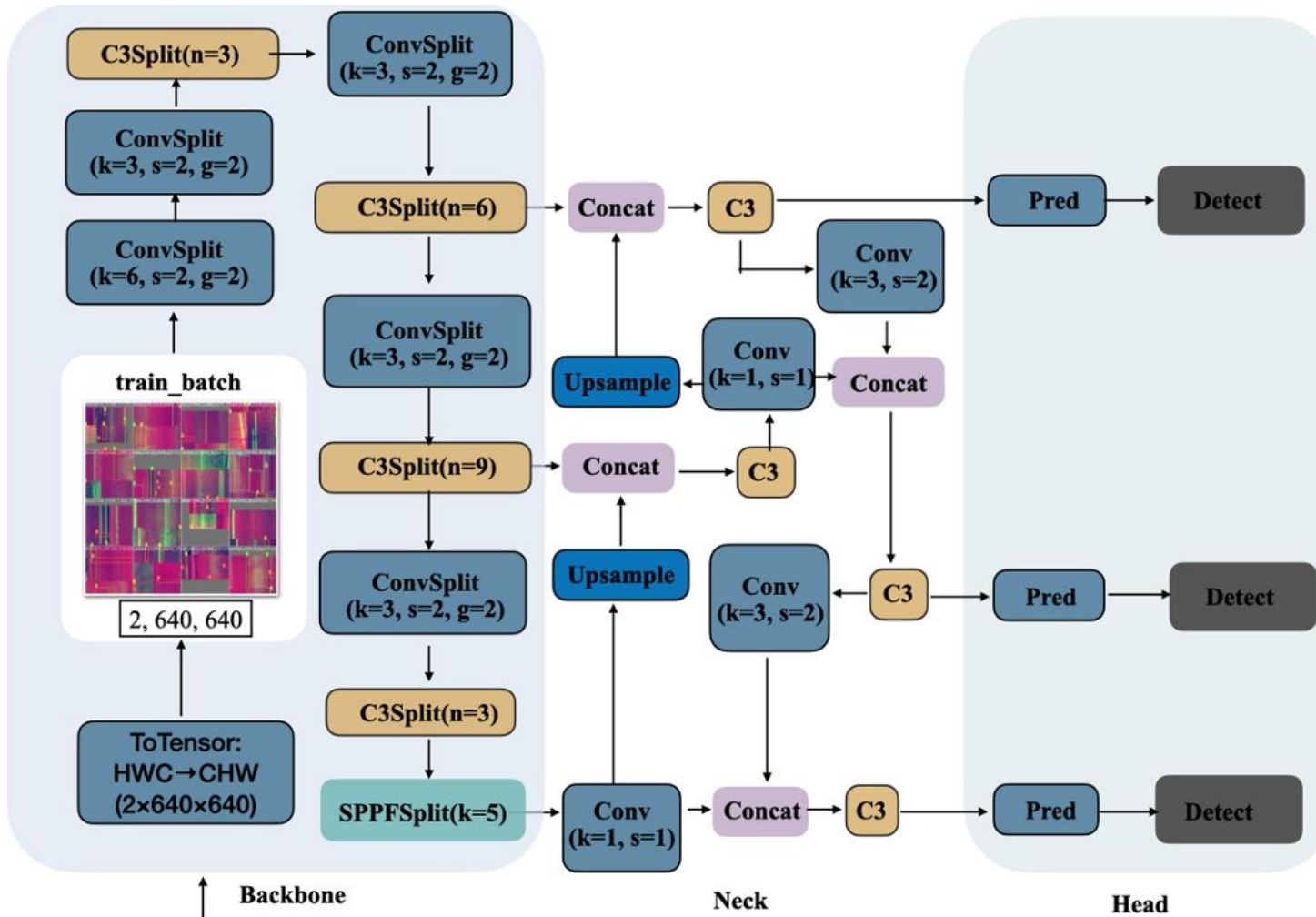


Figure 5: Split-backbone YOLOv5s architecture for intensity-range pavement distress detection.

Key design:

- Only the backbone is modified.
- Backbone Conv, C3, and SPPF blocks are replaced with ConvSplit, C3Split, and SPPFSplit.
- Intensity and range streams remain separated in the backbone.
- Neck and head follow the standard YOLOv5s design.
- Fusion happens after backbone feature extraction.

Backbone Components: ConvSplit, C3Split, and SPPFSplit

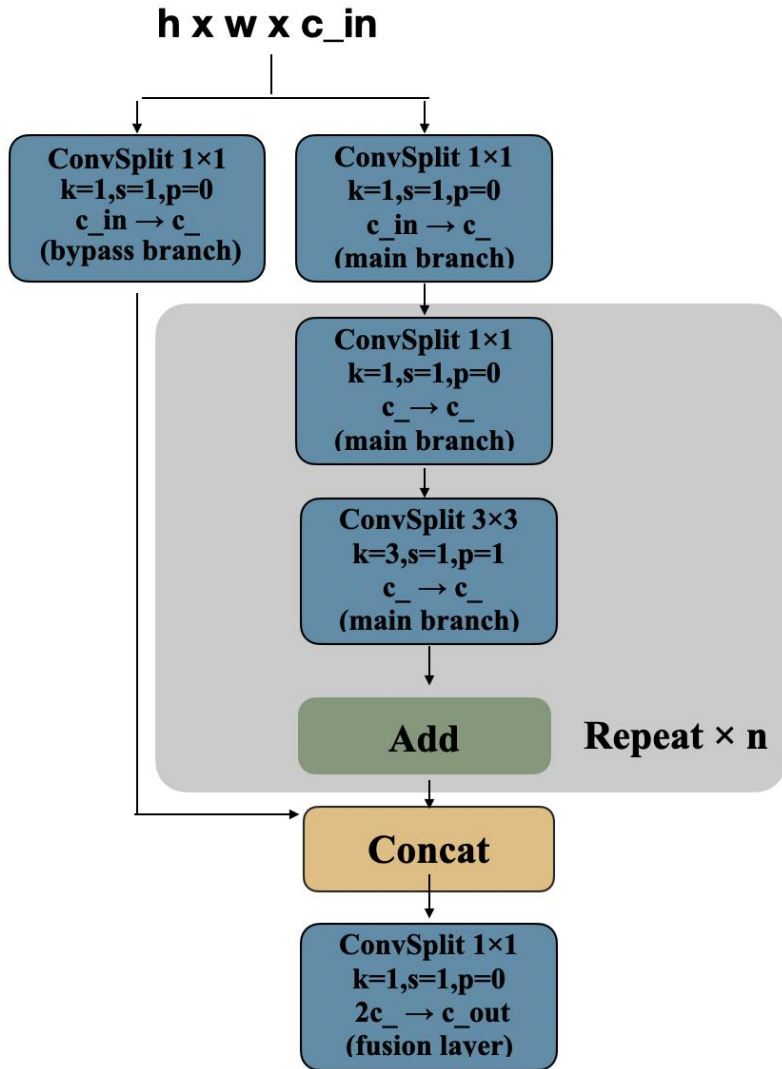


Figure 6: C3Split module replacing standard C3 convolutions with ConvSplit operations.

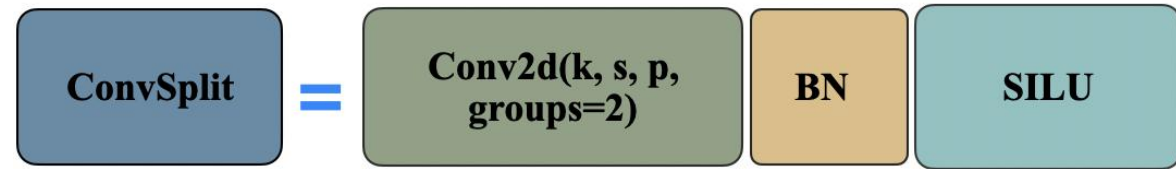


Figure 7: ConvSplit module implemented by grouped convolution with two modality groups.

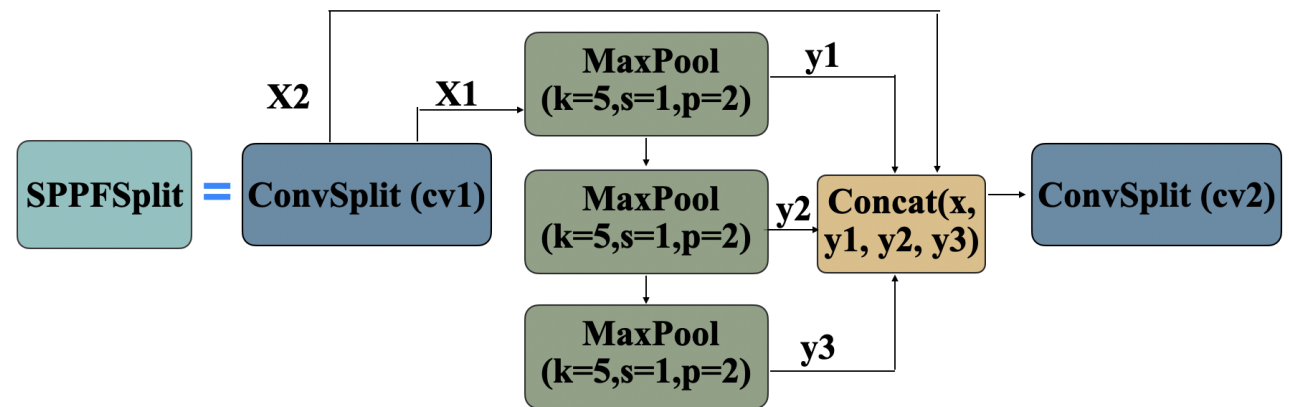
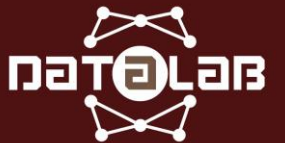


Figure 8: SPPFSplit module preserving multi-scale pooling while using ConvSplit projections.

Experimental Setup



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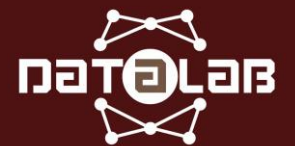
Training Setup

Table 2: Training and evaluation setup for intensity-range YOLOv5s detection.

Item	Setting
Task	ACP pavement distress detection
Dataset	TxDOT intensity-range ACP dataset
Models	YOLOv5s variants
Baselines	I-only, R-only, Early Fusion
Proposed	Split-backbone YOLOv5s + designed augmentation
Input size	640 × 640
Epochs	250
Metrics	AP@0.5 / AP@0.5:0.95



Results



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Test Set Results

Key Findings on Test Set

- Proposed method remains effective on the held-out test set.
- Early Fusion 2ch performs worse, so the gain is not from simply reducing channels.
- Split Backbone improves performance by delaying modality fusion.
- Designed augmentation performs best by using class- and modality-aware training.

Table 3: Overall performance on the held-out TxDOT ACP test set.

Method	AP@0.5	AP@0.5:0.95	Precision	Recall
Early Fusion 3ch	0.775	0.391	0.781	0.724
Early Fusion 2ch	0.736	0.367	0.782	0.667
Split Backbone 2ch	0.799	0.402	0.781	0.754
Split Backbone + Uniform Aug	0.811	0.412	0.775	0.763
Split Backbone + Designed Aug	0.833	0.428	0.770	0.785

Class-wise Modality Preference

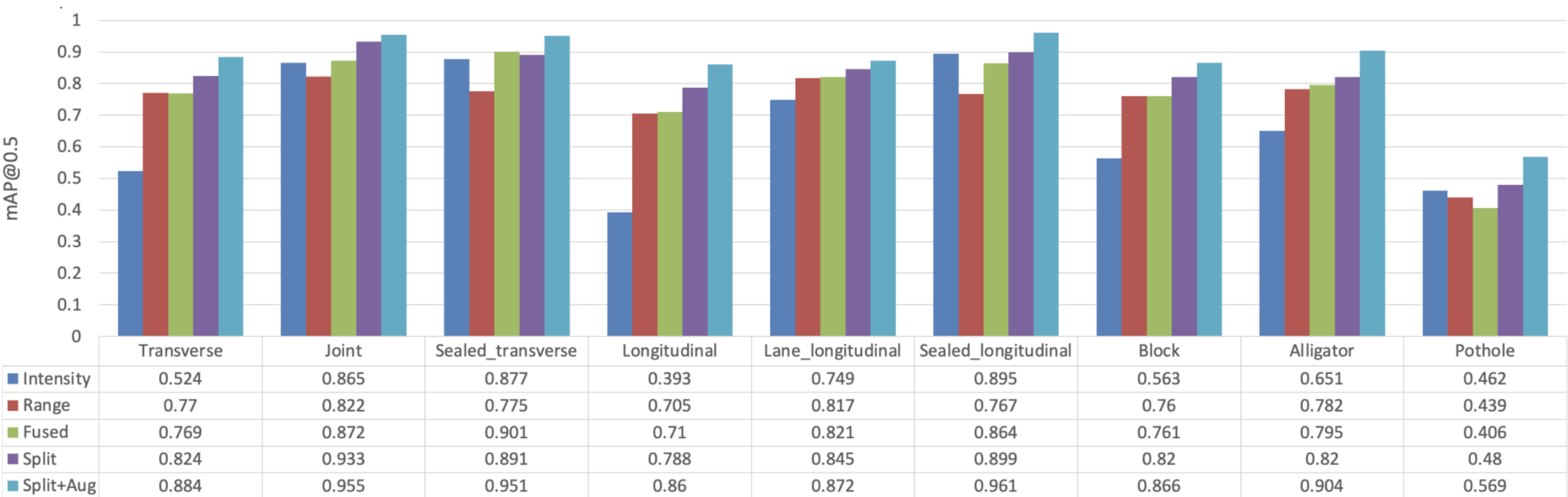


Figure 6: Class-wise AP@0.5 comparison under intensity-only, range-only, early-fusion, split-backbone, and proposed settings.

Qualitative Results

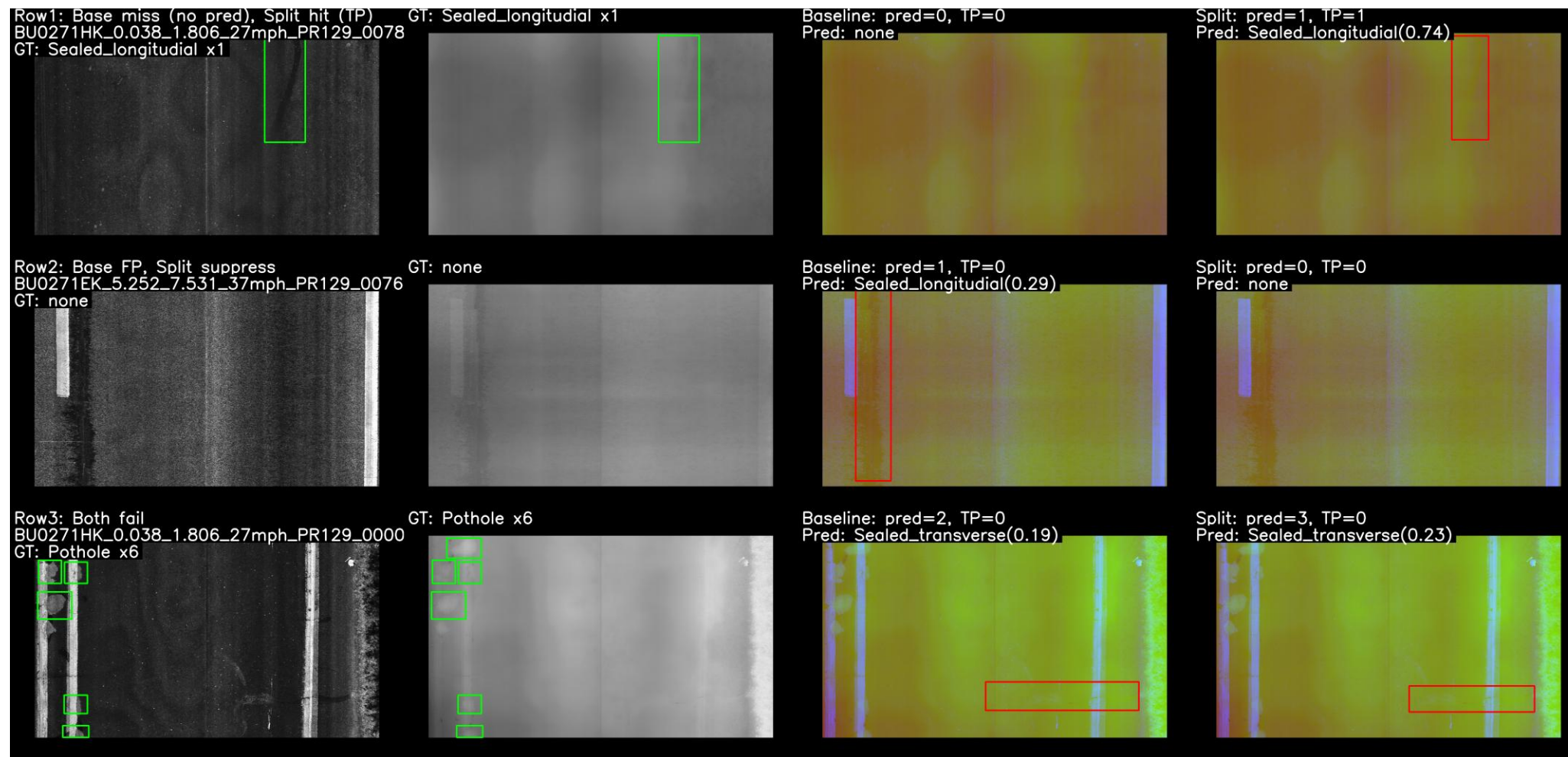


Figure 7: Qualitative examples comparing intensity image, range image, early-fusion baseline prediction, and proposed model prediction.

Conclusion / What I Tried and Lessons Learned



Conclusion: Key Findings

- Early fusion is simple, but it can mix modality signals too early.
- Intensity and range provide complementary but class-dependent evidence.
- Split-backbone learning helps preserve modality-specific features.
- Modality-aware augmentation improves robustness by respecting the physical meaning of each channel.
- The best performance comes from combining architecture design and data-centric augmentation.

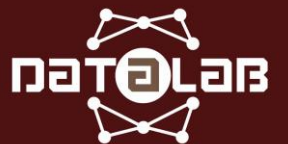


What I Learned from This Independent Study

- Multimodal learning requires more than simply stacking input channels.
- Data preprocessing and augmentation must respect the physical meaning of each modality.
- Controlled ablation studies are necessary to verify where improvements come from.
- Model complexity and real inference speed do not always follow the same trend.
- The final work was developed into an ACM ICMR 2026 accepted paper.



Thank you. Questions?



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